

General Notes

- Following erection notes are very important for the safety and proper erection of the building. The building must be erected according to the erection drawings and safe erection practices. And it should be approved by a qualified engineer.
- Mammut Building System is not responsible for any structural inadequacy or any other problems if the above requirements not followed by the customer.**
- Mammut always provides one set of complete bill of materials (for all the components) to the customer, along with the erection drawings to facilitate checking and verification of all the materials received at site are as per the requirement.
- Customer / Erector must read the complete set of erection drawings and general erection notes given below prior to start of any erection activities.
- Customer / Erector should ensure and confirm that the latest revision of the drawing and documents are referred.
- Mammut's dedicated team of professional customer- service engineers are always at your assistance.

Fabrication

- Fabrication shall be in accordance with Mammut standard practices in compliance with the applicable sections relating to design requirements and allowable stresses of the latest edition of the "AWS Structural Welding Code D1.1".
- Fabrication tolerances are as per section 9 of Metal Building Systems Manual (MBMA 2006)

Primer

- Shop primer paint is a rust inhibitive primer which meets the end performance of SSPC. This primer is not intended for long term exposure to the elements. Mammut is not responsible for any deterioration of the shop primer paint as a result of improper handling and/or storage.
- Mammut shall not be responsible for any field applied paint and/or coatings. (Section 6.5 AISC Code of Standard Practice, 13th Edition).

Bolt Tightening

- All bolted connections used by Mammut are considered to be in the non-slip-critical category and therefore need only be tightened to the Snug Tight Condition. The Snug Tight Condition is defined as the tightness that exists when all plies in a joint are in firm contact. This may be attained by a few impacts of an impact wrench or the full effort of a person using an ordinary spud wrench.
- Galvanized High Strength Bolts to be lubricated and free from dust / debris. And Galvanized nut with lubricant provided must be related to a snug tight condition without stripping. If the Erector noticed any lack of such lubrication, he is required to lubricate them at site, before tightening.
- It is the erectors responsibility to use the correct specified bolts, wrong usage may cause failure of the building.

Errors and Repairs

- Claims for correction of alleged mistakes will be disallowed unless Mammut has received prior notice thereof and allowed reasonable inspection of such mistakes. The correction of minor mistakes by the use of drift pins to draw the components into line, moderate amounts of reaming, chipping and cutting, and the replacement of minor storages of material are a normal part of erection and are not subject to claim. No part of the building may be returned for alleged mistakes without the prior approval of Mammut. (MBMA Section 6.10)

Doors and Windows

- Mammut assumes that all roll up doors and windows, personnel doors which are not supplied by Mammut, have been designed to resist the required wind load of your building. In many cases, openings which have not been designed for full wind load will reclassify the building as partially open, thereby increasing wind coefficients. Increased wind coefficients may require additional girts and purlins, as well as increased wind reactions of the rigid frames. If you are using or planning to use a roll up doors, or any other material to enclose your building which has not been designed for full wind load, notify Mammut in advance so that your building can be properly designed for partially open conditions.

Bracing

- Erecting permanent bracing and temporarily guying the structure during erection prior to placing sheathing / cladding is extremely important. The removal of permanent bracing and / or providing insufficient temporary guy lines may cause the collapse of the building during or after erection which will be the Contractor's / Owner's responsibility.
- All bracing strapping and briding shown and provided by Mammut for this building is required and shall be installed by the erector as a permanent part of the structure. If additional bracing is required for stability during erection, it shall be the erector's responsibility to determine the amount of such bracing and to procure and install as needed.
- Bracing is to be installed as per the erection drawings and must remain tight during the life of the building. Do not remove any bracing without the prior written consent of the Engineering Department at Mammut Building Systems.

Unloading Material at Site

- Unloading of material at site is the responsibility of the customer / contractor / erector, due our policy Mammut drivers may not assist in the unloading of material.
- Inspections
- Field inspections and inspection costs are the responsibility of the client unless it is specifically mentioned in the contract.

Material Specification

Important Note:

Subject to market conditions and material availability, Mammut reserves the right to substitute equivalent or better material.

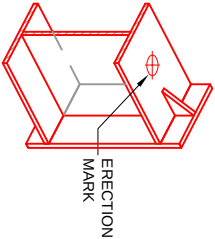
No.	Structural Components	Material Specifications	Strength
1	Built-up Sections	ASTM A 572M Grade 345 OR EN 10025 S355 JR	$F_y = 345 \text{ N/mm}^2$
	Tubes	ASTM A 500M Grade B OR AISI G 3445 ST3M	$F_y = 347 \text{ N/mm}^2$
2	Hot Rolled Sections	ASTM A 572M Grade 345 OR EN 10025 S355 JR	$F_y = 345 \text{ N/mm}^2$
	Channels	ASTM A 572M Grade 345 OR EN 10025 S355 JR	$F_y = 345 \text{ N/mm}^2$
3	Cold formed secondary members	ASTM A 653M SS Grade 340 Class 1 Coating Z 275	$F_y = 340 \text{ N/mm}^2$
4	Sheeting Panels	ASTM A 792M Grade 340 Class 1 Coating AZ1150 Aluzinc coated steel Bare & Pre-painted Aluminum Mill finish & Pre-painted $F_u = 170 \text{ N/mm}^2$ (min)	$F_y = 340 \text{ N/mm}^2$ $F_y = 145 \text{ N/mm}^2$
5	Mezzanine Deck Panels	ASTM A 653M SS Grade 340 Class 1 Coating Z180	$F_y = 340 \text{ N/mm}^2$
6	Diagonal 'X' Bracing	ASTM A 475 - Extra High Strength Class A ($D_t = 2.50$) Rods $F_u = 420 \text{ N/mm}^2$ HR A angles $ASTM - A 572M$ Grade 345 ($F_u = 450 \text{ N/mm}^2$)	$P_u = 120 \text{ kN}$ $F_y = 250 \text{ N/mm}^2$ $F_y = 345 \text{ N/mm}^2$
7	Anchor Bolts	ASTM F 1554 Grade 36 OR DIN 934 Grade 4.6 Hot Dip Galvanized to A193, Class C or equivalent ASTM A325M Type 1 or equivalent (Fully threaded)	$F_y = 250 \text{ N/mm}^2$
8	High Strength Bolts	Hot Dip Galvanized to ASTM A193M Class C ($F_u = 830 \text{ N/mm}^2$) ASTM A 39M OR DIN 933 Grade 4.6	$F_y = 680 \text{ N/mm}^2$
9	Sag Rods	Electro galvanized to ASTM B633	$F_y = 250 \text{ N/mm}^2$
F_y = Yield Strength F_u = Ultimate Tensile Strength F_b = Minimum Breaking Load			

Erection Sequence

- Start erection at the braced bay.
- Used temporary guy wires and ratchet pullers for stability.
- Install permanent bracing i.e. cables, rods, angles, and flange braces as shown in the erection drawings.
- Guy wires should be attached to an immovable object such as cast in anchor bolts, or heavy object.
- Use high strength bolts (HSS) where indicated.
- Pump & align the first bay before proceeding to other bays.
- Tighten all high strength bolts after final alignment of the building.
- All Erection tolerances shall be as per MBMA 2006.
- Sidewall and eave/val girts are likely to sag before sheeting is placed, use temporary supports to align girts before fixing sheets.
- Do not apply any live loads to purlins before roof sheeting is complete.

Erection Marks

When steel members have critical orientation requirements and it may be possible that these member are fit incorrectly, an Erection Mark can be found on the member (as shown in the right side sketch). This Erection Mark is a hard stamped marking and is always on the top of the member and indicated in the erection drawings by the same symbol.



BUYER / END USE CUSTOMER Responsibilities

- It is the responsibility of the BUYER / END USE CUSTOMER to obtain appropriate approvals and secure necessary permits from City, District, State or Federal Agencies as required, and to advise / release Mammut to fabricate upon receiving such information.
- Mammut standard specifications apply unless stipulated otherwise in the Contract Documents. Mammut's design, fabrication, quality criteria, standards, practice, methods and tolerances shall govern the work with any other interpretations to the contrary not withstanding. It is understood by both Parties that the BUYER / END USE CUSTOMER is responsible for clarification of inclusions or exclusions from the architectural plans and / or specifications.
- In case of discrepancies between Mammut's structural drawings and other trades, Mammut's structural plans govern. (Section 3 AISC Code of Standard Practices, 13th Edition).
- Approval of Mammut drawings and calculations indicates that Mammut has correctly interpreted and applied the Contract Documents. The approval constitutes the contractor / owners acceptance of the Mammut's design concepts, assumptions, and loading. (Section 4 AISC Code and MBMA 3.3.3).
- Once the BUYER / END USE CUSTOMER has signed Mammut's Approval Package and the project is released for fabrication, changes shall be billed to the BUYER / END USE CUSTOMER including material, engineering, and other costs. An additional fee may be charged if the project must be moved from the fabrication and shipping schedule.
- The BUYER / END USE CUSTOMER is responsible for overall project coordination. All interface, compatibility, and design considerations concerning any materials not furnished by Mammut and materials furnished by Mammut are to be considered and coordinated by the BUYER / END USE CUSTOMER. Specific design criteria concerning this interface between materials must be furnished before release for fabrication or Mammut's assumptions will govern (Section 4 and Commentary AISC Code of Standard Practice, 13th Edition).
- It is the responsibility of the BUYER / END USE CUSTOMER to ensure that Mammut's structural drawings comply with applicable requirements of any governing building authorities. The supplying of sealed engineering data and drawings for the metal building system does not imply or constitute an agreement that Mammut or its design engineers are acting as the engineer of record or design professional for a construction project. These drawings only to certify the design of the structural components furnished by Mammut.
- Mammut is responsible for the design of the anchor bolts to permit the transfer of force between the base plate and anchor bolts in shear, bearing and tension, but is not responsible for the transfer of anchor bolts forces to concrete or the adequacy of the anchor bolts in relation to the concrete. Unless otherwise provided in the Order Documents, Mammut does not design and is not responsible for the design, material and construction of the foundation or foundation embedments. The BUYER / END USE CUSTOMER should assure himself that adequate provisions are made in the foundation design for loads imposed by column reactions of the building, other imposed loads, and bearing capacity of the soil and other conditions of the building site.
- It is recommended that the anchorage and foundation of the building be designed by a Qualified Civil Engineer experienced in the design of such structures. (Section A3.1 MBMA 2006 Metal Building Systems Manual).
- Normal erection operations include the corrections of minor mistakes by moderate amounts of reaming, chipping, welding or cutting, and the drawing of elements into line through the use of drift pins. Errors which cannot be corrected by the foregoing means or which require major changes in member configuration are to be reported immediately to Mammut by the BUYER / END USE CUSTOMER to enable whoever is responsible either to correct the error or to approve the most efficient and economical method of correction to be used by others. (Section 7 AISC Code of Standard Practices, 13th Edition).
- Neither Mammut nor the BUYER / END USE CUSTOMER will cut, drill or otherwise alter the work, or the work of other trades, to accommodate other trades, unless such work is clearly specified in the contract documents. Whenever such work is specified, the BUYER / END USE CUSTOMER is responsible for furnishing complete information as to materials, size, location and number of alterations prior to preparation of shop drawings. (Section 7 AISC Code of Standard Practice, 13th Edition).
- WARNING:** In no case should Aluzinc steel panels be used in conjunction with lead or copper. Both lead and copper have harmful corrosive effects on the Aluzinc alloy coating when they are in contact with Aluzinc steel panels. Even run-off from copper flashing, wiring or tubing onto Aluminized Zinc should be avoided.
- SAFETY COMMITMENT:** Mammut has a commitment to manufacture quality building components that can be safely erected. However, the safety commitment and job site practices of the erector are beyond the control of Mammut. It is strongly recommended that safe working conditions and accident prevention practices be the top priority of any job site. Local, District, State and Federal safety and health standards should always be followed to help insure worker safety. Make certain all employees know the safest and most productive way of erecting a building. Emergency procedures should be known to all employees. Daily meetings highlighting safety procedures are also recommended. The use of hard hats, rubber sole shoes for roof work, proper equipment for handling material, and safety nets where appropriate, are recommended.

CARBON STEEL SCREENS FOR SINGLE SKIN PANEL									
SDS-A-820	SDS-5-825	SDS-5-840	SDS-5-860	SDS-5-880	SDS-A-820	SDS-5-820	POP RIVET	BEAD MASTIC	FOAM CLOSURE
SDS9001	SDS9002	SDS9003	SDS9004	SDS9009	SDS9016	SDS9016	SPW4016	SPW4012	SPW4008 / SPW4009
DRAWING SYMBOL									
ACTUAL DRAWING									
SPW4008									
SPW4009									

JOB NUMBER		BUILDING NO.		DRAWING NO.	
DACS-036		B01		N01	
DRAWING TITLE					
ERECTOR NOTES					
3108000100					

0	05.02.16	ISSUED FOR CONSTRUCTION	XJB	PMH	BPS
NO.	DATE	DESCRIPTION	DSM	DPH	CHK

PROJECT: WAREHOUSE
LAHORE - PAKISTAN

CUSTOMER: COLGATE PALMOLIVE (PAKISTAN) LIMITED
LAKSON SQUARE, BUILDING NO. 2,
SAFARW SHAHEED ROAD, KARACHI-74200, PAKISTAN

STEEL CONTRACTOR:

P.O. Box 4188
HF-2, Shigar - U.A.E.
E-mail : sales@mammut.ae
Website : www.mammut.ae
Head Office & Pakistan
Phone : 9971 6154 9999
Fax : +971 6154 9993